

Filename: cisco-ccna200301-1-7-1-ip_addressing_ipv6_basics_and_address_types

ShowName: Cisco CCNA (200-301)

Topic: 1.0 Networking Fundamentals

EpisodeName: IP Addressing: IPv6 Basics & Address Types

Description: In this episode, Aubri and Ronnie look into the growing use of IPv6. They discuss the basics of identifying IPv6 addresses and why we should be using them.

IP Addressing: IPv6 Basics & Address Types

1.9 Compare IPv6 address types

Basic

- Compare with IPv4 (32bit vs 128bit)
- Analogies
- Format: Hexadecimal e.g. 2001:1fbc:2cca:3ddb:a123:b234:c345:d456/64

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Code starts
+ 1.9.a Global unicast
+ 2000::/3 - 3fff::/3
+ Routable IPv6 address on the internet = IPv4 public address
+ Devices can be configured
+ Manually configured
+ Stateless Address Autoconfiguration
+ Stateful DHCPv6
+ 1.9.b Unique local
+ FC00::/7 - FDFF::/7
+ Equivalent to IPv4 private addresses
+ 1.9.c Link local
+ FE80::/10 - (FEBF::/10)
+ Unique only on it's own subnet
+ not forwarded by routers
+ only one link-local address per interface
+ Device Configuration
+ Can create own link-local address when device start
+ Can be manually configured
+ Role:
+ Used to communicate with other devices on network
+ A router's link local address is the default gateway for other addresses
+ used to exchange IPv6 dynamic routing information (messages and routing
information)
+ 1.9.d Anycast
+ Can be assigned to to more than one device.
+ routed to the "nearest" device with that anycast address according to
routing table of the router.
+ no special prefix but uses same as global unicast
+ 1.9.e Multicast
+ FF00::/8
+ One to Many
+ Well known multicasts:
+ FF02::1 - all IPv6 devices
+ FF02::2 - all IPv6 routers
+ FF02::5 - all OSPFv3 routers
+ FF02::A - all EIGRP (IPv6) routers
+ 1.9.f Modified EUI 64
+ Refer to notes EUI-64
Code ends
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